Report on Data Modeling project

Performance Comparison of Feedforward Neural Network, Convolutional Neural Network, and Recurrent Neural Network on the Breast Cancer Dataset

Executed by

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1. Introduction

The Breast Cancer dataset from the UCI Machine Learning Repository is widely used for binary classification problems. The dataset consists of 569 samples and 30 numerical features, such as mean radius, texture, perimeter, area, smoothness, and more, to help predict whether a given tumor is malignant or benign. In this report, we compare the performance of three neural network architectures — **Feedforward Neural Network (FNN)**, **Convolutional Neural Network (CNN)**, and **Recurrent Neural Network (RNN)** on this dataset. The objective is to identify which model achieves the best classification performance and to understand how each model architecture handles the dataset.

2. Dataset Overview

The Breast Cancer dataset includes 30 input features and a target variable that denotes whether a tumor is **malignant (0)** or **benign (1)**. The dataset contains slightly more benign tumors (63%) than malignant ones (37%), but this imbalance is manageable and does not require resampling techniques.

Basic Statistics of the Dataset:

- **Mean radius**: 14.13 ± 3.52
- **Mean texture**: 19.29 ± 4.30
- **Mean perimeter**: 91.97 ± 24.30
- **Mean area**: 654.89 ± 351.91

3. Data Preprocessing

To prepare the dataset for each neural network architecture, the following preprocessing steps were carried out:

1. **Standardization**: All features were scaled to have zero mean and unit variance, which is critical for neural networks to ensure faster convergence during training.
2. **Data Splitting**: The data was split into training (80%) and test sets (20%).
3. **Reshaping**:
    - For the **FNN**, the data remained in its original shape of (455, 30) for the training set.
    - For the **CNN**, the data was reshaped to (455, 30, 1) to simulate a 1D image where each sample has 30 "channels" (features).
    - For the **RNN**, the data was reshaped to (455, 1, 30), treating each sample as a sequence of 30 timesteps.

4. Model Architectures and Training

4.1. Feedforward Neural Network (FNN)

- **Architecture**:
  - Input layer with 30 neurons (one per feature).
  - Two hidden layers with 64 and 32 neurons, each using the ReLU activation function.
  - Output layer with a single neuron using the sigmoid activation function (for binary classification).
- **Training**: The FNN model was trained for 20 epochs using the Adam optimizer and binary cross-entropy loss.

4.2. Convolutional Neural Network (CNN)

- **Architecture**:
  - A 1D convolutional layer with 64 filters, followed by a flattening layer.
  - Dense hidden layers with 32 neurons.
  - Output layer with a single neuron using the sigmoid activation function.
- **Training**: The CNN model was trained for 20 epochs under the same conditions as the FNN model. The input data was reshaped to (455, 30, 1) to allow for convolution over the 30 features.

4.3. Recurrent Neural Network (RNN)

- **Architecture**:
  - An LSTM (Long Short-Term Memory) layer with 64 units, which is particularly suited for handling sequential data.
  - A Dense layer with 32 neurons.
  - Output layer with a single neuron using the sigmoid activation function.
- **Training**: The RNN was also trained for 20 epochs. The input data was reshaped to (455, 1, 30) to treat each sample as a sequence of 30 timesteps, even though the features are not naturally sequential.

5. Results and Discussion

5.1. Loss Curves

During training, the loss curves were plotted for both the training and validation sets to monitor model performance and overfitting.

- **FNN**: The FNN model converged quickly, with the loss decreasing consistently over epochs. However, the gap between the training and validation loss suggested some overfitting, particularly after epoch 15.
- **CNN**: The CNN model showed similar behavior, though it had a slightly more stable validation loss curve, indicating better generalization compared to FNN.
- **RNN**: The RNN model had the most volatile loss curve. The model showed some difficulty in converging during the early epochs but stabilized by the end of the training.

5.2. Model Evaluation

After training, each model was evaluated on the test set, and the classification metrics were generated, including **precision**, **recall**, **F1-score**, and **support**.

FNN Classification Report:

	precision	recall	f1-score	support
0	0.94	0.88	0.91	42
1	0.94	0.97	0.96	72
accuracy			0.94	114
macro avg	0.94	0.93	0.93	114
weighted avg	0.94	0.94	0.94	114

CNN Classification Report:

	precision	recall	f1-score	support
0	0.93	0.90	0.92	42
1	0.96	0.97	0.97	72
accuracy			0.95	114
macro avg	0.94	0.94	0.94	114
weighted avg	0.95	0.95	0.95	114

RNN Classification Report:

	precision	recall	f1-score	support
0	0.92	0.90	0.91	42
1	0.96	0.97	0.96	72
accuracy			0.95	114
macro avg	0.94	0.94	0.94	114
weighted avg	0.95	0.95	0.95	114

5.3. Performance Comparison

- **Accuracy**: Both the CNN and RNN achieved an accuracy of **95%**, slightly outperforming the FNN, which had an accuracy of **94%**.
- **Precision and Recall**: For both classes (malignant and benign), the CNN and RNN demonstrated superior precision and recall compared to the FNN. The CNN showed slightly better generalization for the benign class (class 1).
- **F1-score**: The RNN and CNN achieved marginally higher F1-scores compared to the FNN. This indicates a better balance between precision and recall in both architectures.

6. Conclusion

The results of this experiment show that both **CNN** and **RNN** models slightly outperform the traditional **FNN** in terms of classification accuracy, precision, recall, and F1-score on the Breast Cancer dataset. Despite the fact that CNNs are typically used for image-based data and RNNs are generally employed for sequential data, both models demonstrate their ability to generalize well even on structured, tabular datasets like this one.

In terms of practicality, the FNN offers a simpler and faster training process, while CNN and RNN provide better performance and generalization. For real-world applications where precision and recall are crucial, such as medical diagnosis, the **CNN** and **RNN** architectures may be more suitable due to their slightly superior predictive abilities. However, the choice of model should also consider factors such as computational complexity and the available data structure.

7. Future Work

Future investigations could include:

- Experimenting with more advanced CNN and RNN architectures.
- Performing hyperparameter optimization using techniques like grid search.
- Investigating the effects of dataset imbalance by applying resampling methods.
- Exploring other neural network architectures, such as Transformer models, which have shown great success in various domains.